

3.8 Interconnect of peripherals

Several peripherals have direct connections between them. This allows autonomous communication between peripherals, saving CPU resources thus power supply consumption. In addition, these hardware connections allow fast and predictable latency.

Depending on peripherals, these interconnections can operate in Run, Sleep and Stop modes.

Table 4. Interconnect of peripherals

Interconnect source	Interconnect destination	Interconnect action	Run	Sleep	Stop
TIMx	TIMx	Timer synchronization or chaining	Y	Y	-
	ADCx	Conversion triggers	Y	Y	-
	DMA	Memory-to-memory transfer trigger	Y	Y	-
ADCx	TIM1	Timer triggered by analog watchdog	Y	Y	-
RTC	TIM16	Timer input channel from RTC events	Y	Y	-
All clock sources (internal and external)	TIM14,16,17	Clock source used as input channel for RC measurement and trimming	Y	Y	-
CSS RAM (parity error)	TIM1,16,17	Timer break	Y	Y	-
CPU (hard fault)	TIM1,16,17	Timer break	Y	-	-
GPIO	TIM1,3	External trigger	Y	Y	-
	ADC	Conversion external trigger	Y	Y	-

3.9 Clocks and startup

The clock controller distributes the clocks coming from different oscillators to the core and the peripherals. It also manages clock gating for low-power modes and ensures clock robustness. It features:

- **Clock prescaler:** to get the best trade-off between speed and current consumption, the clock frequency to the CPU and peripherals can be adjusted by a programmable prescaler
- **Safe clock switching:** clock sources can be changed safely on the fly in run mode through a configuration register.
- **Clock management:** to reduce power consumption, the clock controller can stop the clock to the core, individual peripherals or memory.